

MACHINE LEARNING-BASED ANOMALY DETECTION FOR TRAFFIC IN IoT-ENABLED TRANSPORTATION NETWORKS

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Overview

Connected vehicles exchange real-time data across V2V, V2I, and satellite channels — expanding the attack surface and necessitating robust network intrusion detection. [1].



Fig. 1: Connected transportation network illustrating V2V, V2I, and V2S wireless communication channels across vehicles, roadside units, and GPS satellites.

Dataset: ACI-IoT-2023

- 1,231,406 network flow records, 78 features
- 11 attack classes: Port Scan (35.8%), Benign (26.7%), ICMP Flood (18.3%), DNS Flood, Vulnerability Scan, Ping Sweep, OS Scan, Slowloris, SYN Flood, Dictionary Attack, UDP Flood
- Rare classes (<100 samples) removed; stratified downsample to 500K records
- 70/15/15 train/val/test split; StandardScaler fit on training set only (no data leakage)

Feature Importance

Three features account for 79.65% of all classification decisions (Random Forest Gini importances). Only packet headers need to be inspected — no deep packet inspection required, enabling lightweight real-time deployment on V2X edge devices [2, 1].

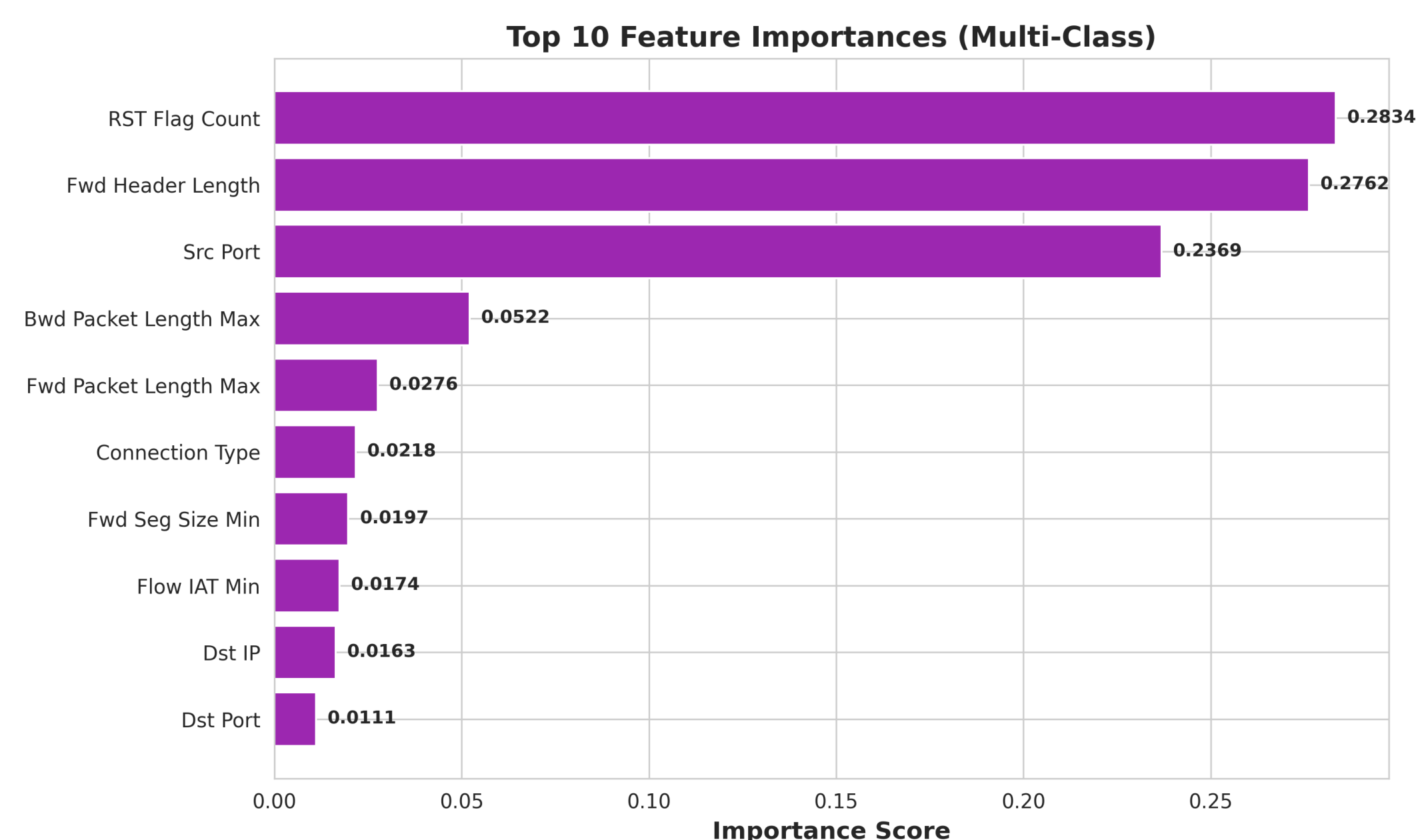


Fig. 2: Top 10 most important features by Random Forest Gini impurity importance.

Binary Classification Results

All classical models achieved >99.5% accuracy [3, 4]. XGBoost delivered the best AUC (0.9999) and fastest training time (0.08 min). The Decision Tree produced the fewest false positives (74), minimizing false alarms in deployment.

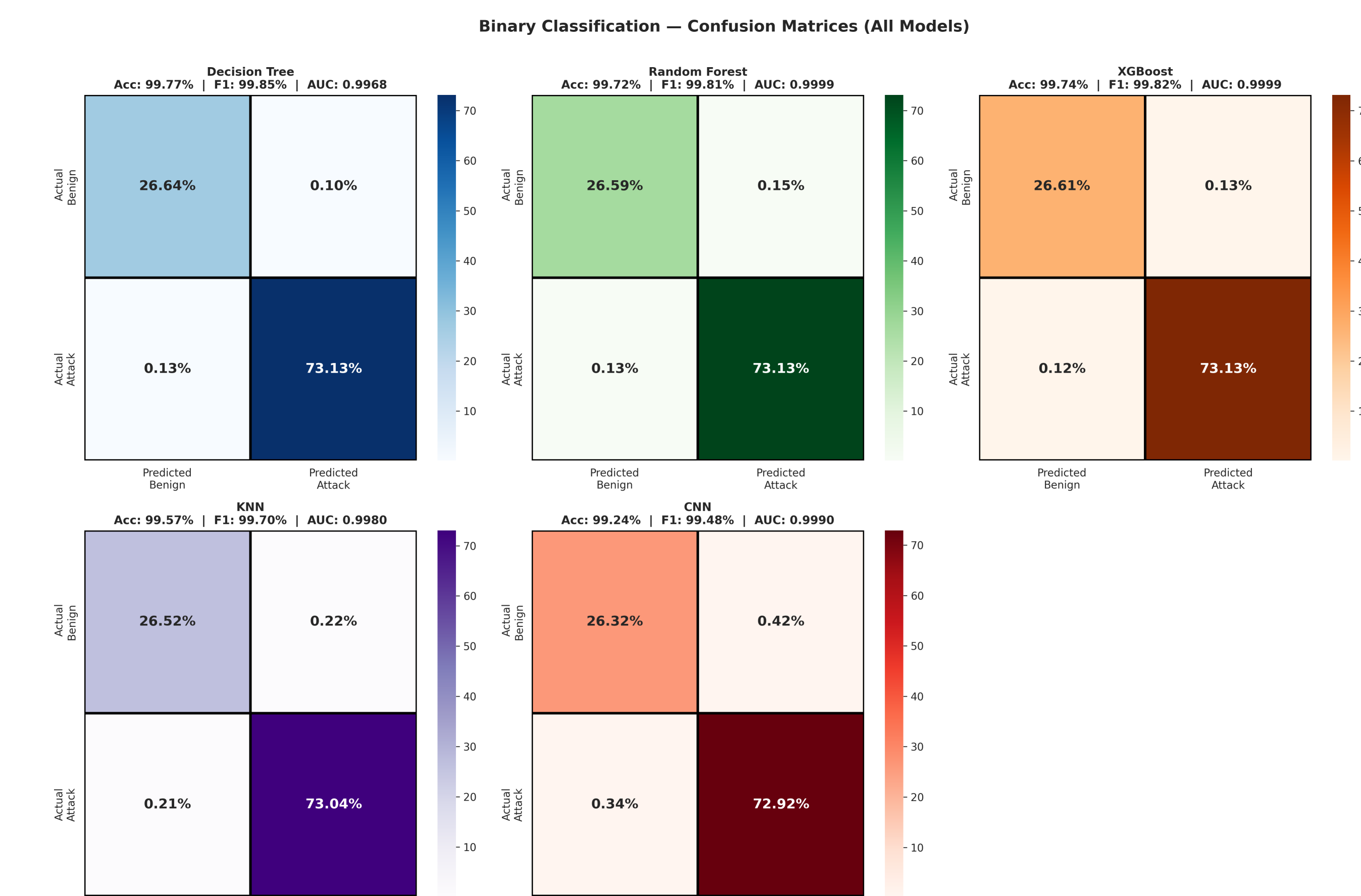


Fig. 3: Confusion matrices for all five models on the binary classification task (Benign vs. Attack).

Multi-Class Classification Results

The single Decision Tree outperformed all models — including ensembles and the CNN — in 11-class classification [5, 3]. Distinct attack signatures favor clean decision boundaries over ensemble voting or convolutional feature extraction.

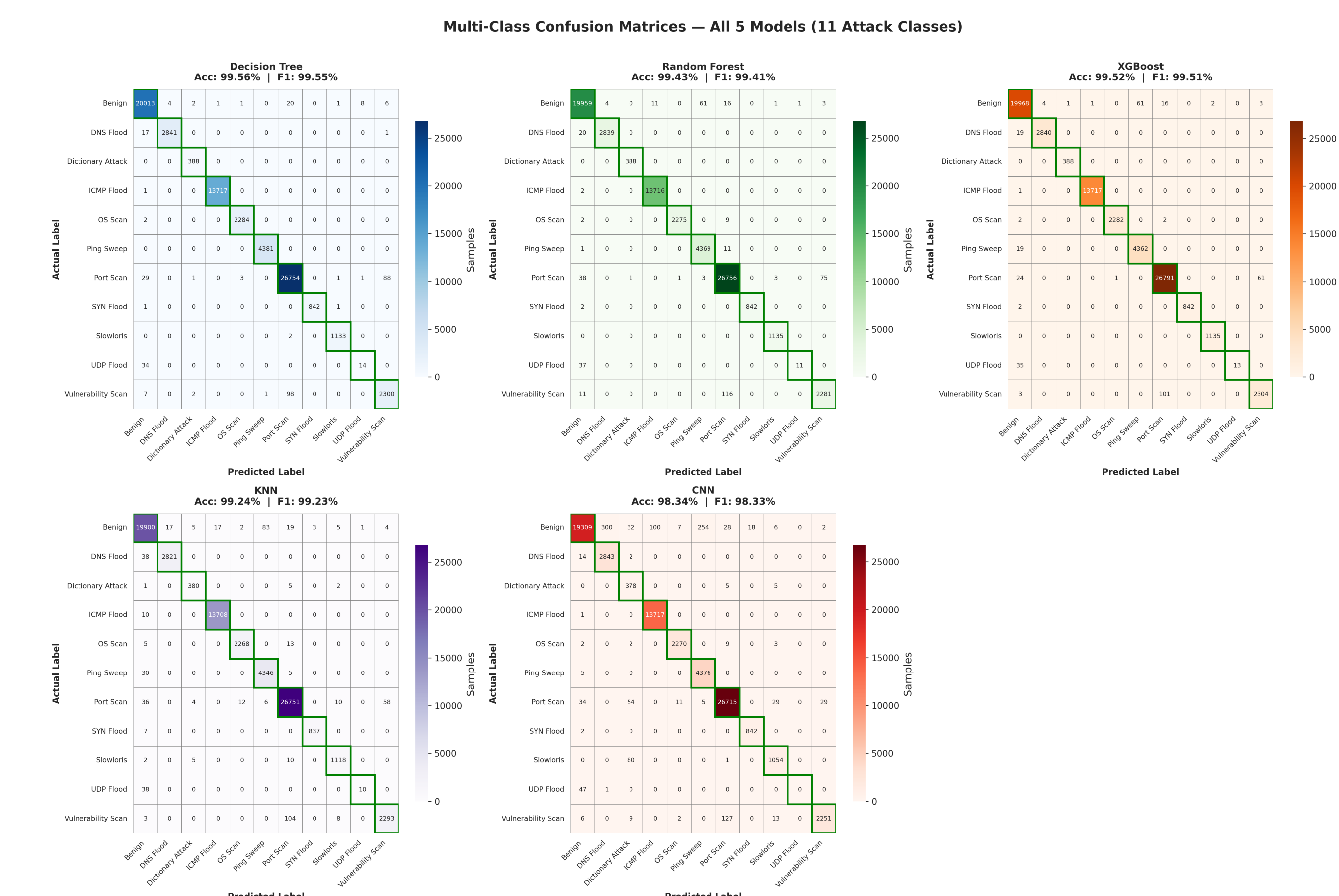


Fig. 4: Confusion matrices for all five models on the 11-class attack classification task.

Key Findings & Conclusion

- Classical ML models consistently **outperformed deep learning** for tabular network intrusion detection; the CNN (98.34% F1) ranked last across all five models.
- **XGBoost** is recommended for production deployment: best AUC (0.9999) and fastest training time (0.08 min).
- **Decision Tree** excelled in 11-class classification due to sharp, deterministic decision boundaries, and is well-suited for resource-constrained V2X edge nodes.
- All classical models achieved **>99.5% accuracy** detecting cyber threats in IoT transportation networks [2, 1, 3, 4].
- **Header-only inspection** (no deep packet inspection) enables lightweight real-time protection on V2X edge devices [2, 1].
- Three features alone account for **79.65%** of all classification decisions, confirming that compact models are sufficient for deployment.

Model Performance Comparison

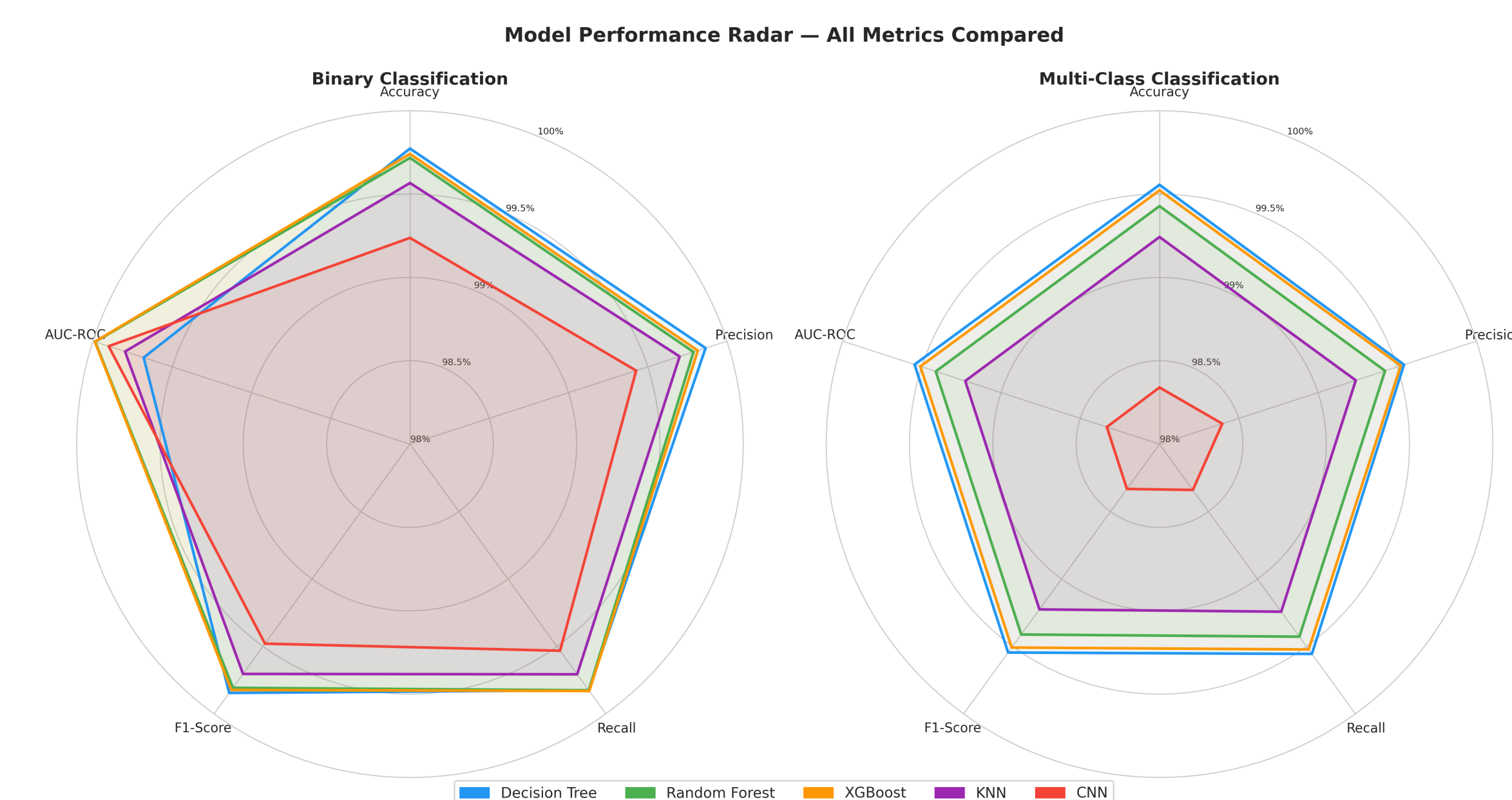


Fig. 5: Radar charts comparing all five models across Accuracy, Precision, Recall, F1-Score, and AUC-ROC for binary (left) and multi-class (right) classification.



GitHub Repository



Research Website



Connor's LinkedIn



Molly's LinkedIn